

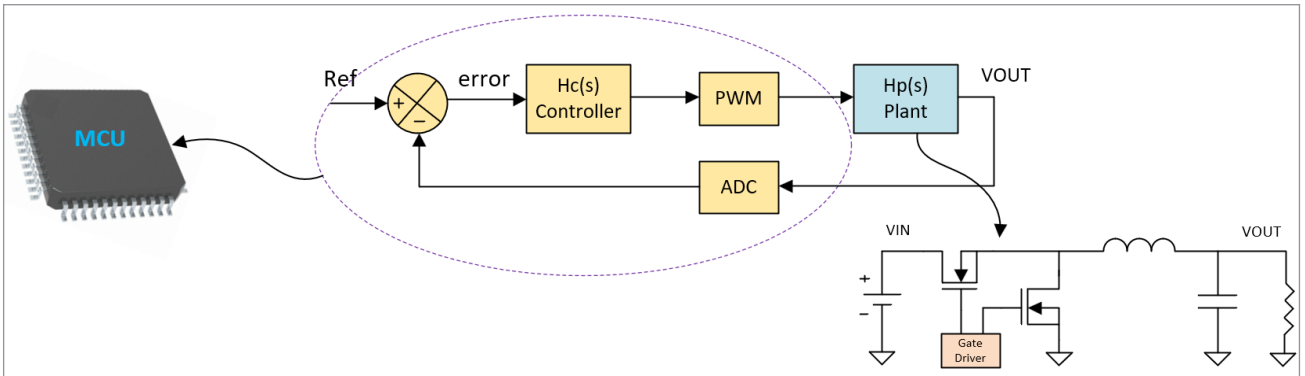


Fig. 6: Voltage mode control in digital power supply system schematic (Credit: Biricha Digital)

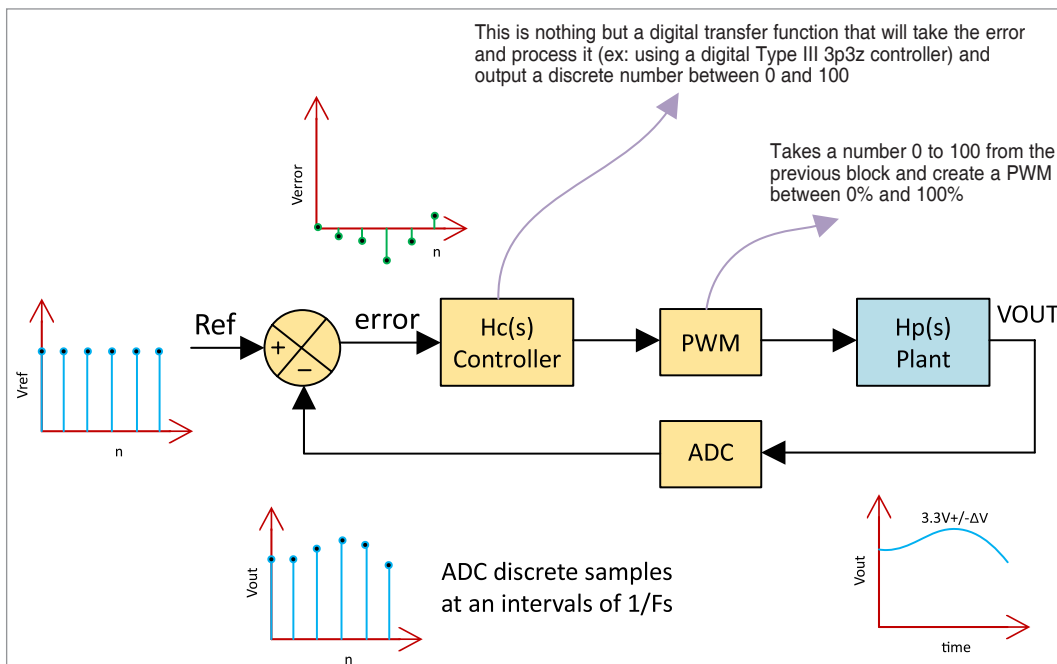


Fig. 7: The digital controller serving as a digital equivalent of an op-amp-based compensator (Credit: Biricha Digital)

positioned to achieve a desirable gain and phase margin. This shapes the frequency response of the power supply to meet stability criteria.

Voltage mode control (digital PSU)

In digital power supplies, the functionality largely mirrors that of analogue PSUs. However, instead of a voltage divider and operational amplifier (op-amp) creating the compensator, the process is executed

digitally within the microcontroller.

The output voltage (Vout) is a continuous-time signal sampled by the microcontroller ADC. Inside the microcontroller, the ADC measures the voltage against a reference voltage. The difference between these values produces an error signal processed by a digital controller. The digital controller applies a linear difference equation (Type III or Type II controller) to calculate a discrete value representing the PWM duty cycle (0% to 100%).

The digital controller functions

- Single feedback loop
- Double pole compensation
- Slower dynamic response

Peak current mode control (PCMC).

- Built-in current limiting
- Faster response to input voltage changes
- Two feedback loops
- Single pole compensation
- Slope compensation is required
- Blanking is necessary to handle current spikes

Typically, there are two types of compensators.

Type II compensator.

- Commonly used for current mode control

- Two poles and one zero

Type III compensator.

- Widely used for voltage mode control
- Three poles and two zeros

Voltage mode control (analogue PSU)

By selecting appropriate values for passive components (R & C) in the compensator network (also called the controller), poles and zeros can be